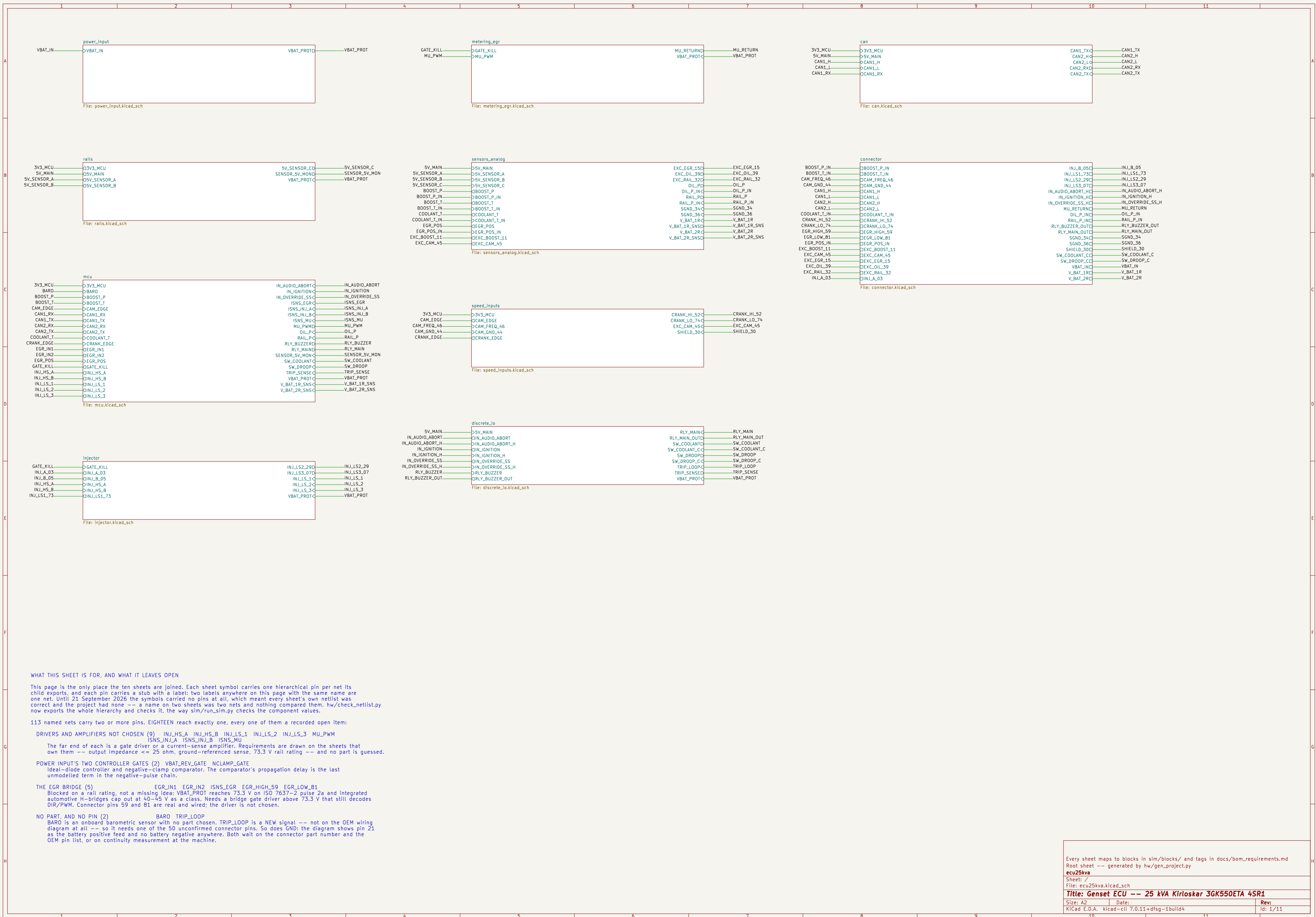
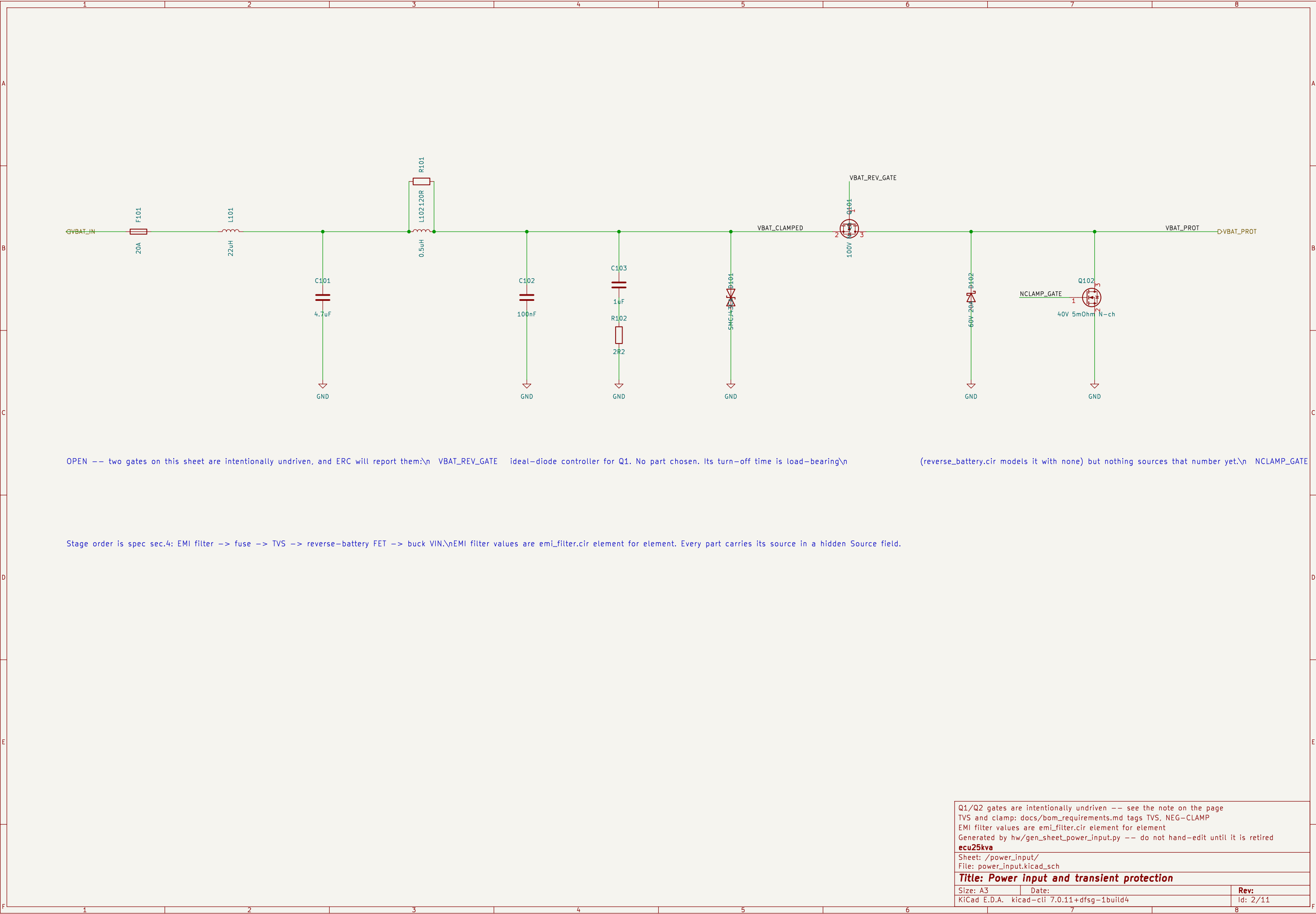






OPEN -- two gates on this sheet are intentionally undriven, and ERC will report them:\n VBAT_REV_GATE ideal-diode controller for Q1. No part chosen. Its turn-off time is load-bearing\n (reverse_battery.cir models it with none) but nothing sources that number yet.\n NCLAMP_GATE comp

Stage order is spec sec.4: EMI filter -> fuse -> TVS -> reverse-battery FET -> buck VIN.\nEMI filter values are emi_filter.cir element for element. Every part carries its source in a hidden Source field.

Q1/Q2 gates are intentionally undriven -- see the note on the page

TVS and clamp: docs/bom_requirements.md tags TVS, NEG-CLAMP

EMI filter values are emi_filter.cir element for element

Generated by hw/gen_sheet_power_input.py -- do not hand-edit until it is retired

ecu25kva

Sheet: /power_input/

File: power_input.kicad_sch

Title: Power input and transient protection

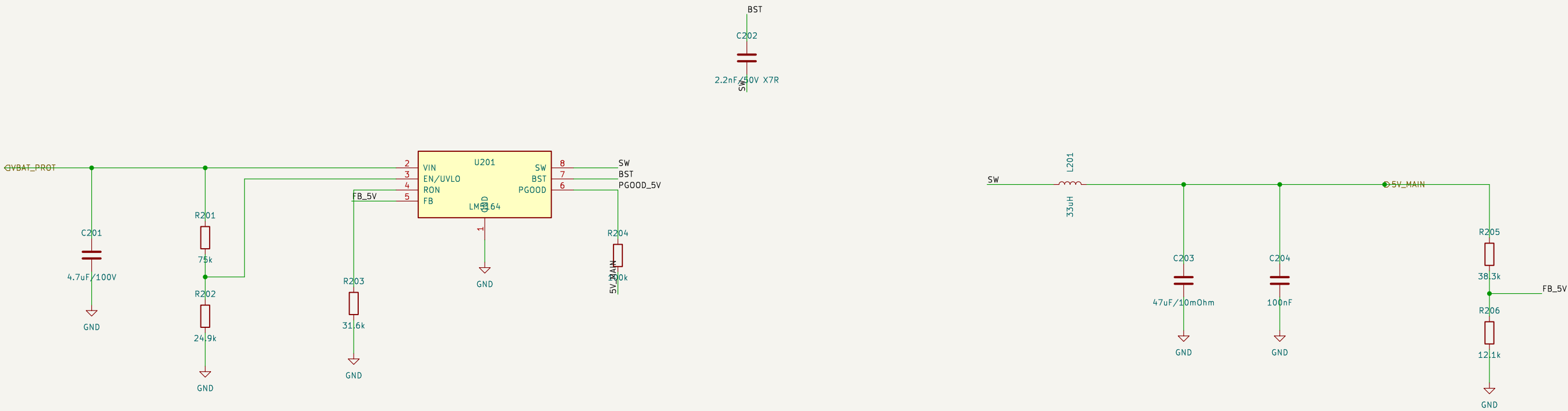
Size: A3

Date:

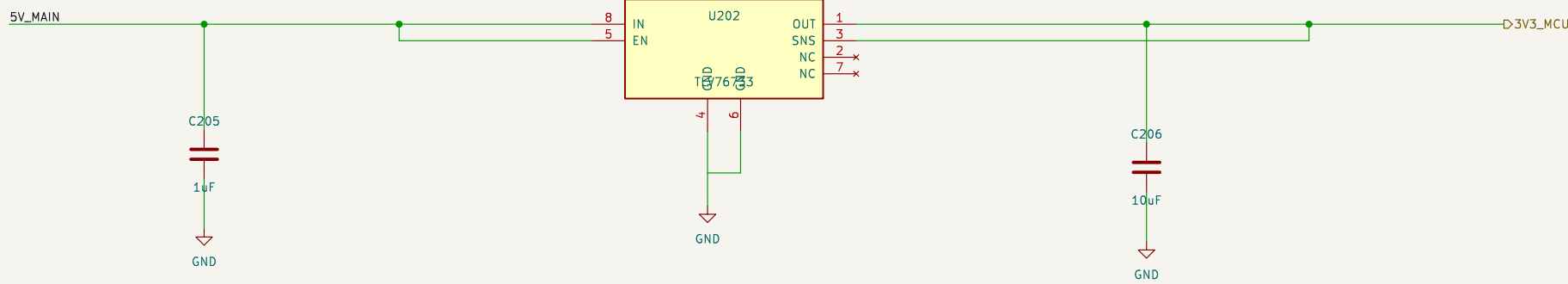
Rev:

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

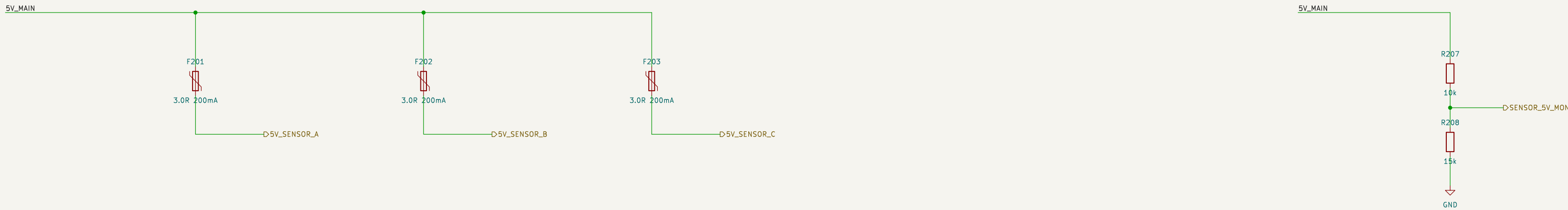
Id: 2/11



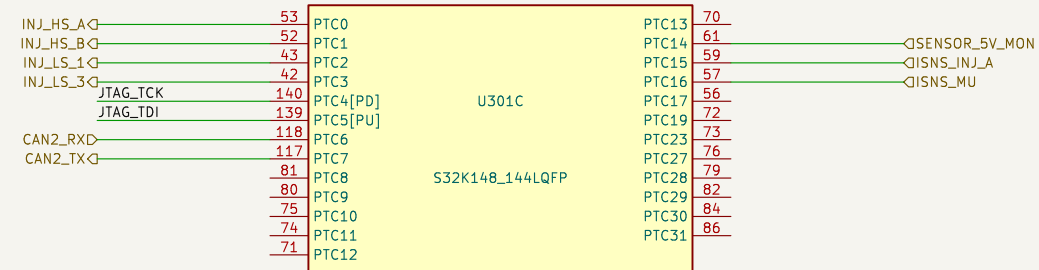
TWO PARTS CHOSEN HERE, both replacing research memo 07 entries that were marked representative rather than chosen:
LM5164 memo 07 sec.4 names a 60 V TPS54360B-Q1. transient_clamp.cir now carries a PASSING check that reads "60 V-class parts are NO LONGER viable downstream" -- raising the TVS standoff to 43 V, so the part stops conducting during a NORMAL clamped dump, put the pulse 2a clamp at 73.3 V. Every rating check in the suite has been written against the LM5164's own -0.3 V / 100 V VIN limits all along; the BOM was simply never reconciled with it. A 60 V buck here is a part that fails the first pulse 2a event.
TLV76733 memo 07 sec.6 names TLV1117-33QDCYRQ1, explicitly "class pricing, not fetched". The supervisor chosen on the mcu sheet made 3V3 accuracy load-bearing: outside 3.143-3.459 V the board resets itself. TLV767-Q1 is 1% over load AND temperature, giving 3.267-3.333 V.



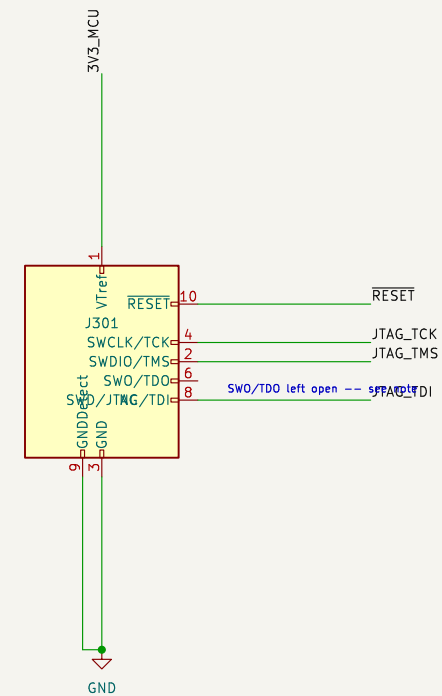
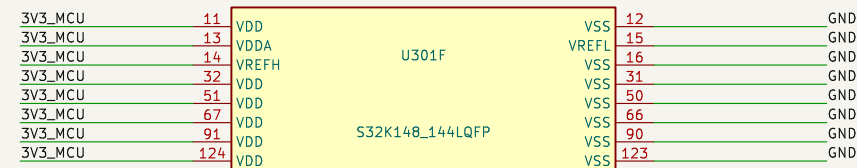
RATIOMETRIC MONITOR, a limitation worth stating: the divider taps 5V_MAIN, upstream of the PTCs. Each group's own PTC drops about 50 mA x 3.0 ohm = 150 mV, which is 3% of the rail and is NOT captured by a measurement taken before it. Correct for the regulator's drift, not for the protection element's. Moving the tap downstream would fix one group and leave the other two wrong, so the tap stays common and the limit is recorded here.



PGOOD_5V is brought out and left unassigned. The LM5164 asserts it open-drain; the pin map has no channel for it. Available, not wired to anything -- shown rather than hidden.



OPEN: SET0/SET1 strap the watchdog window ratio, and the right ratio depends on a firmware loop period that does not exist yet (phase 4). Link resistors, not hard wires.



```

PTIO/PT1E1 are added here -- pinmap.md assigns no watchdog pins
Supervisor IPS3850G33: memo 11 sec.2.1 + TI SBV5301B Table 5-1
Pin assignments: docs/pinmap.md. Pin positions: hw/lib/s32k148_layout.json
Generated by hw/gen_sheet_mcu.py -- do not hand-edit until it is retired
ec25kva

Sheet: /mcu/
File: mcu_sch

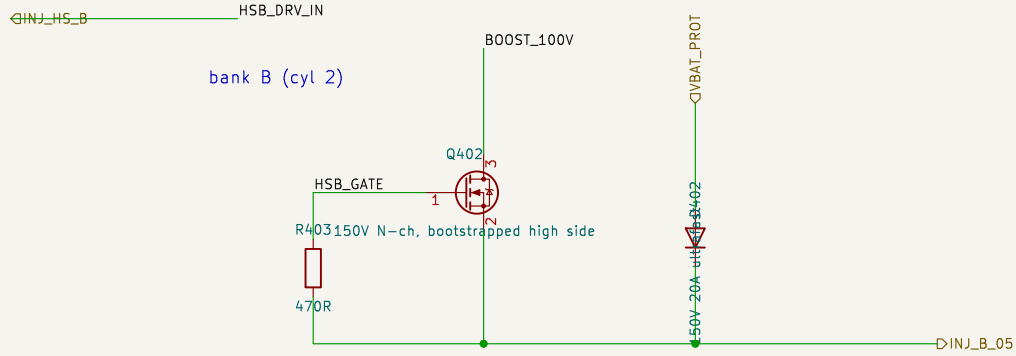
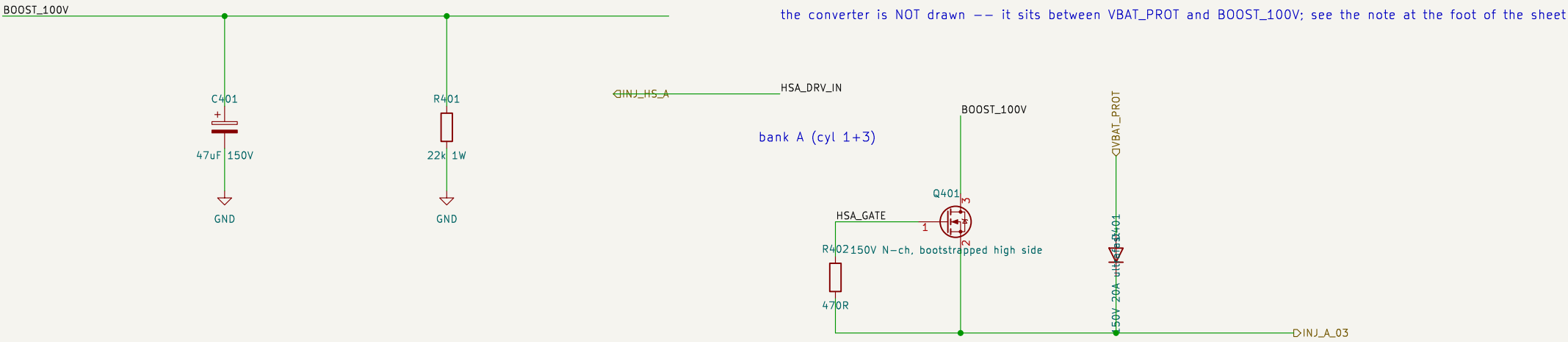
MCU, supervisor and debug

Size: A2      Date:
KiCad E.D.A.  kiCad-1c1 7.0.11+dfsg-1build4

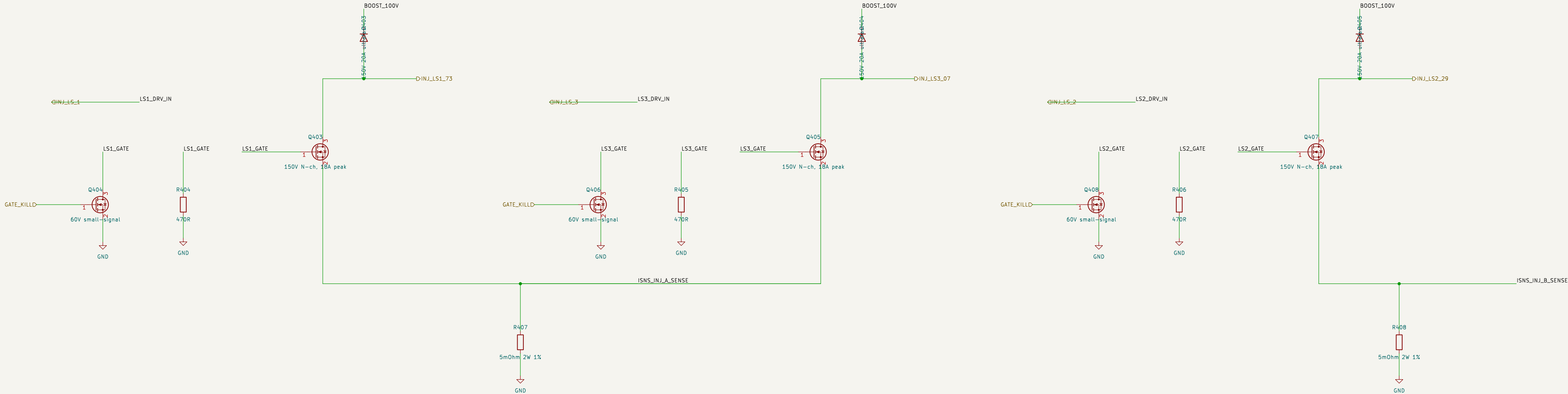
Rev:
Date: 4/11

```

BOOST RAIL -- 100 V reservoir for the injector peak phase



GATE_KILL reaches the two high sides through the driver's own SHUTDOWN input, not through a clamp -- a ground-referenced kill FET cannot short a gate that floats 100 V up. Nothing is drawn for it here because the driver is not chosen; see the note at the foot of the sheet.



A GROUND-REFERENCED KILL CLAMP CANNOT SERVICE A HIGH-SIDE GATE, and spec sec.4's fail-safe list names pins 03/05 alongside the four low-side gates as though it could. The three low sides here get exactly the network supervisor.cir sized and docs/pinmap.md sec.2 confirmed pin by pin: a 470 ohm pulldown to ground, plus a small-signal FET that shorts the gate to ground when GATE_KILL asserts, with no MCU involvement -- which is what makes it work against a hung MCU and not only a reset one.

The two high sides cannot use it. The source of a high-side N-FET swings to the boost rail, so a pulldown to GROUND would hold Vgs at minus the source voltage and fight the driver on every edge, and a kill FET referenced to ground cannot short a gate that is floating 100 V up. What is drawn instead is the 470 ohm across GATE and SOURCE -- the same Miller-coupling divider supervisor.cir sized, just referenced to the terminal it has to be referenced to -- and the kill path moves into the driver: GATE_KILL must reach the high-side driver's own shutdown input. That is a REQUIREMENT ON THE UNCHOSEN PART, recorded against the driver rather than drawn, and it is the sixth finding of the shape this project keeps turning up: a document and an executable file agreeing with each other and both wrong about a case neither modelled.

PARTS NOT CHOSEN ON THIS SHEET. Five gate drivers and two current-sense amplifiers, on the same named-net treatment power_input gives its two controller gates.

LOW-SIDE DRIVERS (LS1/LS2/LS3_DRV_IN -> *_GATE). Output impedance <= 25 ohm, from supervisor.cir claim 2: the 470 ohm pulldown is not a perturbation on the driver, it is the bottom half of a divider with it, and at 100 ohm the gate reaches 8.25 V instead of 10 V. 3.3 V logic in (PTC2/PTC3/PTB4, FTM0_CH2/3/4), ~10 V out. HIGH-SIDE DRIVERS (HSA/HSB_DRV_IN -> HSA/HSB_GATE). Same output impedance, but referenced to a source that swings to 100 V, so bootstrapped or isolated, rated above the rail, AND carrying a shutdown input for GATE_KILL per the note above. Supplied from VBAT_PROT and therefore rated for its 73.3 V worst case. CURRENT-SENSE AMPLIFIERS (ISNS_INJ_A/B_SENSE -> PTC15 / PTD19). Ground-referenced -- that is the whole reason the shunts went low-side -- gain about 33 to put 90 mV at full scale near 3.0 V, and bandwidth enough to resolve a 38 us ramp to peak, so 100 khz or better. A Kelvin tap at the shunt pad, not a via into the ground pour: 90 mV sits on a return carrying 18 A and a few milliohms of pour between pad and reference is the whole signal.

THE BOOST CONVERTER ITSELF IS NOT DRAWN. boost_converter.cir models it as an average charging current and says so: 'no inductor, no switch, no control loop.' That is enough to size the reservoir and it is not enough to draw a converter, so what is on this sheet is the part the block DID size -- the 47 uF reservoir, its 150 V class, and the bleed -- with VBAT_PROT and BOOST_100V naming the two ends of the stage nobody has chosen. What it has to do: 50 mA average from VBAT_PROT into a 100 V setpoint, which recovers the 7.3 V peak-phase droop in 6.8 ms against the 26.67 ms between cylinders at 1500 rpm.

Shunts settled LOW-SIDE and per-bank -- docs/pinmap.md 1.6 asked for the placement
Hold current arrives through a battery DIODE-OR, not a second switched rail
boost_converter.cir / injector_boost.cir / injector_turnoff.cir
Generated by hw/gen_sheet_injector.py -- do not hand-edit until it is retired

ecv25kva		
Sheet: /injector/		
File: injector.kicad_sch		
Title: Injector drive: boost rail, high sides, low sides		
Size: A2	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		
Id: 5/11		



IN5S_MU_SENSE goes to a current-sense amplifier, also not chosen. 50 mOhm at the 0.675 A setpoint is 33.75 mV, measured ACROSS the shunt -- the amplifier's low reference is the shunt's own ground pad, which is electrically GND and so is not a separate net. It IS a layout constraint: a Kelvin tap at the pad, not a via into the ground pour, because 33.75 mV sits on a return carrying up to 1.35 A and a few milliohms of pour between pad and reference is the whole signal.

Output goes to PTC16 (ADCO_SE14). The loop is closed in firmware: PTD10 carries the PWM, PTC16 reads the current back. That is arrangement the pin map already assumes, and it is why metering_unit_pwm.cir models the regulator as hysteretic rather than as a fixed duty.

T1 DRV8873-Q1 was the candidate and it fits the functional requirement exactly: an automotive H-bridge with the PH/EN decoded interface bus requirements: EGR-DRIVER tag asks for -- one input pair that cannot express both legs of a diagonal conducting at once, which is the requirement egr_hbridge.cir's 270 A shoot-through produced. AEC-Q100 grade 1, 10 A peak, per-half-bridge current sense on IPROPI1/IPROPI2 that would have fed PDD22 directly.

Its VM absolute maximum is 40 V (SLVSDY7B sec.6.1).

VBAT_PROT reaches 40.0 V for 400 ms on a suppressed load dump and 73.3 V for about 50 us on ISO 7637-2 pulse 2a. An absolute maximum is absolute: out of spec at the dump with zero margin, and at 2a by 1.8x.

THIS IS A FACT ABOUT THE RAIL, NOT ABOUT ONE PART. It applies to the injector high sides, the relay drivers and this bridge alike. transient_clamp.c now carries it as enforced checks rather than as a sentence in a comment, so the next driver sheet fails the suite instead of rediscovering it.

Consequence: integrated automotive H-bridges and smart switches cap out at 40–45 V as a class and are excluded from anything battery-connected here. The EGR bridge goes discrete, for the same reason memo 12 sent the injector drivers discrete. What it needs is a bridge GATE DRIVER rated above 73.3 V that still decodes DIR/PWM internally. No such part has been checked against a retrieved datasheet, and this project does not guess parts into copper.

Unblocked by: one datasheet. The four bridge FETs, the shunt, the gate pulldowns and the kill transistors are all already specified and will draw the same way the metering unit's do on the left of this sheet.

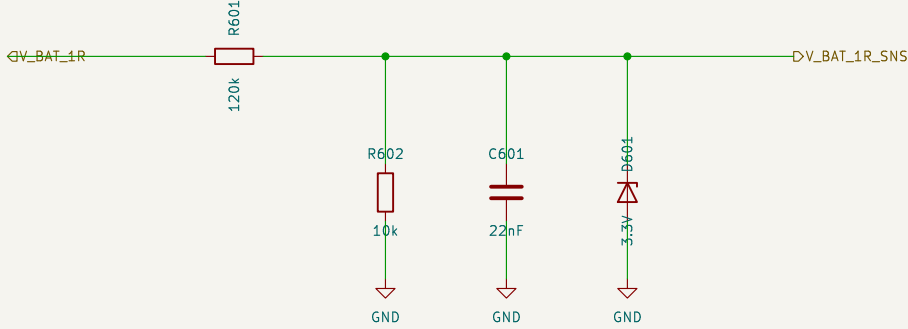
Nothing battery-connected on this board may be rated below 80 V
 EGR half NOT drawn -- blocked on a battery-rail rating, see the note
 Metering unit: metering_unit_pwm.cir + bom_requirements MU=ISENSE, GATE=PULLDOWN
 Generated by hw/gen_sheet_metering_egr.py -- do not hand-edit until it is retired

Sheet: /metering_egr/
File: metering_egr.kicad_sch

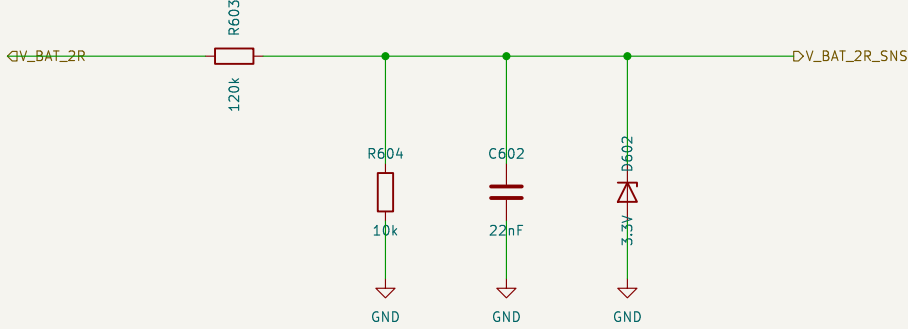
Title: Fuel metering unit and EGR actuator

Size: A2	Date:	Rev:
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 6/11

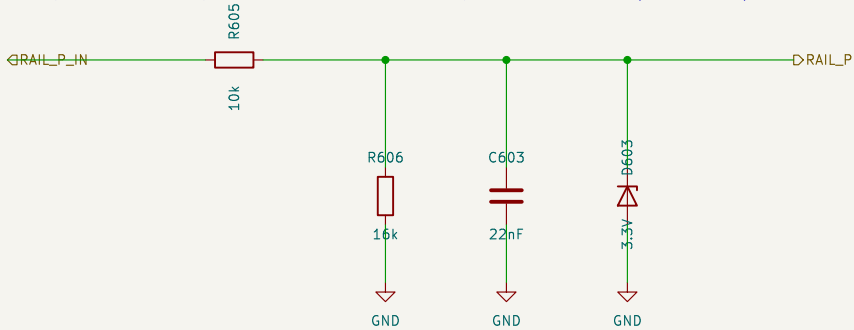
ECU pin 04 -- battery rail 1 -> PTA0 (ADC0_SE0)



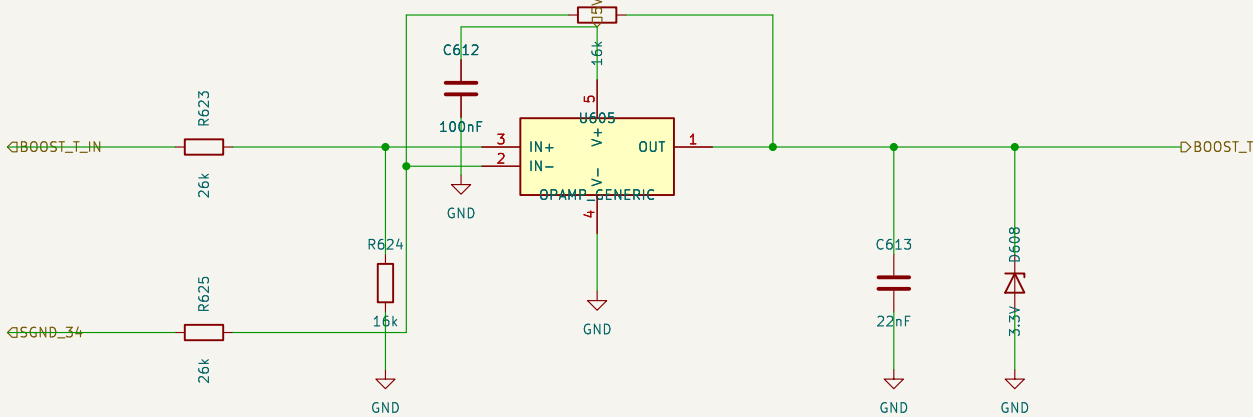
ECU pin 06 -- battery rail 2 -> PTA2 (ADC1_SE0)



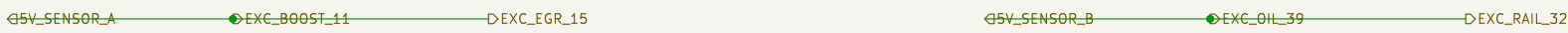
ECU pin 35 -- rail pressure, DEDICATED return pin 08 -> PTA16 (ADC1_SE13)



ECU pin 79 -- boost temperature -> PTA3. U2 OPEN: sensor type unmeasured



5V_SENSOR excitation -- three PTC-protected groups from the rails sheet, out to five connector pins



WHY FIVE CHANNELS ARE DIFFERENTIAL AND ONE IS NOT. The split follows the harness, not the sensor. Boost pressure and coolant temperature SHARE connector pin 34 as their return; EGR position and oil pressure share pin 36. Rail pressure has pin 08 to itself. A shared return carries both sensors' current, so its own resistance appears as an offset on both readings -- sensor_differential.cir puts 20 mA down a 1 ohm corroded return and measures a single-ended reading wrong by the whole offset while the differential one rejects it, a ratio of 90x. A dedicated return does not have that problem, so rail pressure gets the simpler front end.

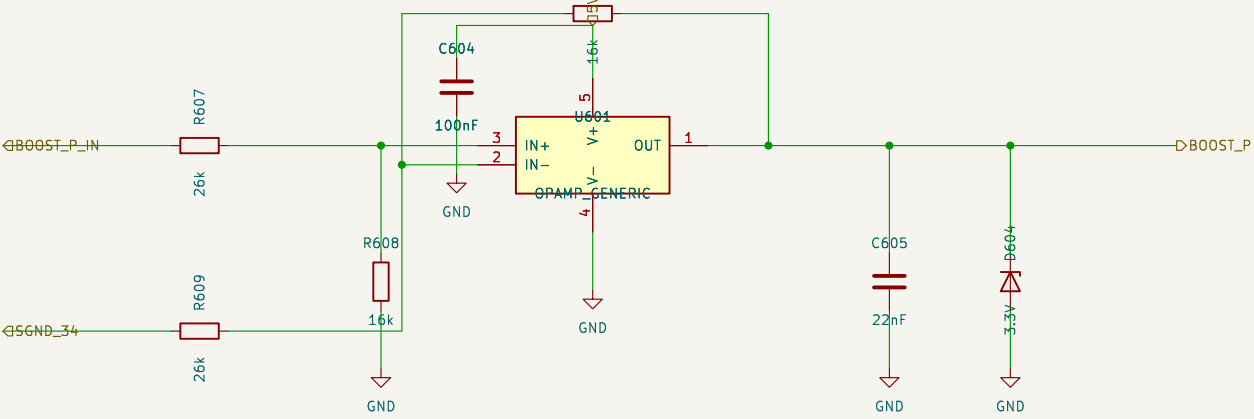
The block states the consequence as a harness requirement, which is the useful form: single-ended is only acceptable if contact resistance is maintained below a figure nobody is going to maintain over a genset's life in an Indian monsoon.

OP-AMPS: FUNCTION DRAWN, PART NOT CHOSEN. OPAMP_GENERIC has placeholder pin numbers and no footprint, deliberately -- placing a real symbol here would assert a choice nobody has made. The requirement IS specified; single 5 V supply, rail-to-rail input and output, and a difference network matched well enough for 60 dB CMRR, which is the floor sensor_differential.cir's own tolerance note tried and found the argument still holds at. The netlist models 80 dB and marks it INFERRED. 0.1% discrete resistors give roughly 48 dB and do NOT clear that floor. A matched network at 0.05% ratio, or a difference amplifier with the network on-die, does -- so this is a resistor-network decision as much as an amplifier one. Five channels is two quad packages.

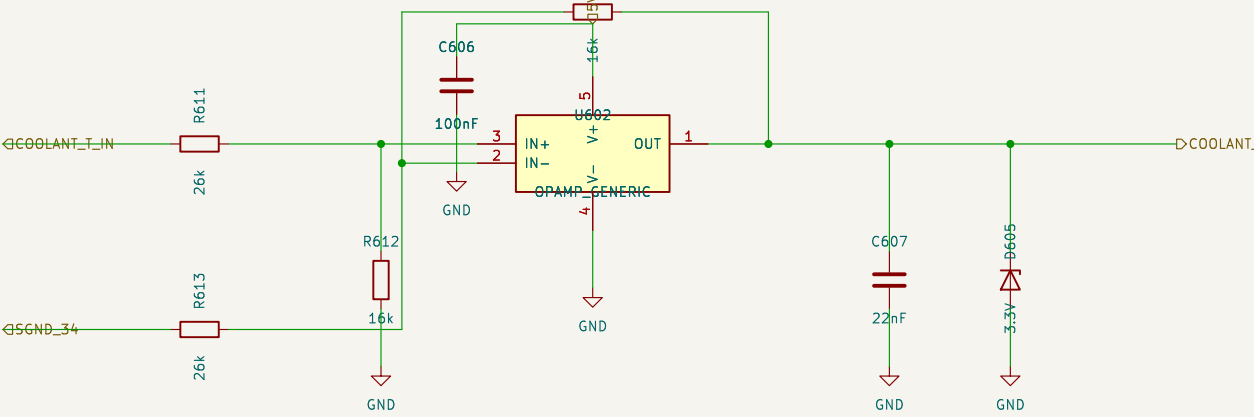
BOOST TEMPERATURE (PTA3, connector pin 79) is drawn differential like its neighbours, but U2 is still open: whether that element is a ratiometric sensor or a bare NTC is unmeasured. ntc_frontend.cir already chose between the two NTC options if it turns out to be one -- a 2.2k pull-up to 5 V, not a 100 uA current source, because the source needs 5.9 V of compliance at -40 C and the rail is 5 V. That is a populate-time change to this channel, not a redesign.

BARO (PTD18) is an onboard sensor, new per spec sec.7 deviation #2. No part is chosen, so nothing is drawn for it -- it needs a device with its own footprint, supply and output type, none of which follow from the pin assignment alone.

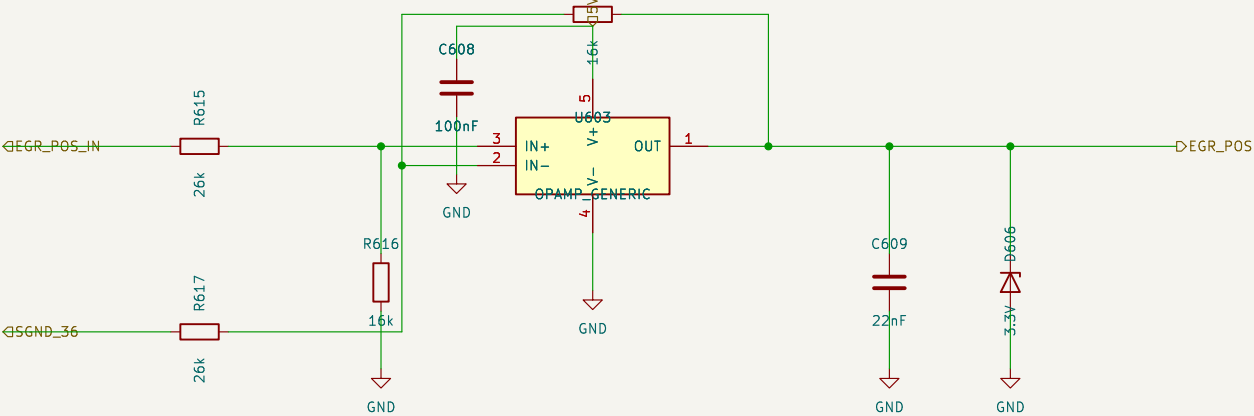
ECU pin 41 -- boost pressure, return SHARED on pin 34 -> PTA1



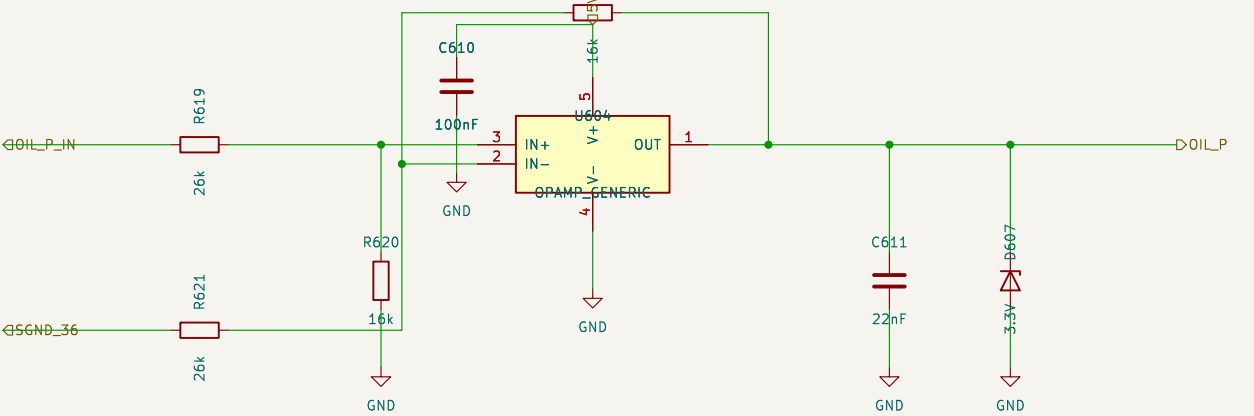
ECU pin 33 -- coolant temperature, return SHARED on pin 34 -> PTA6



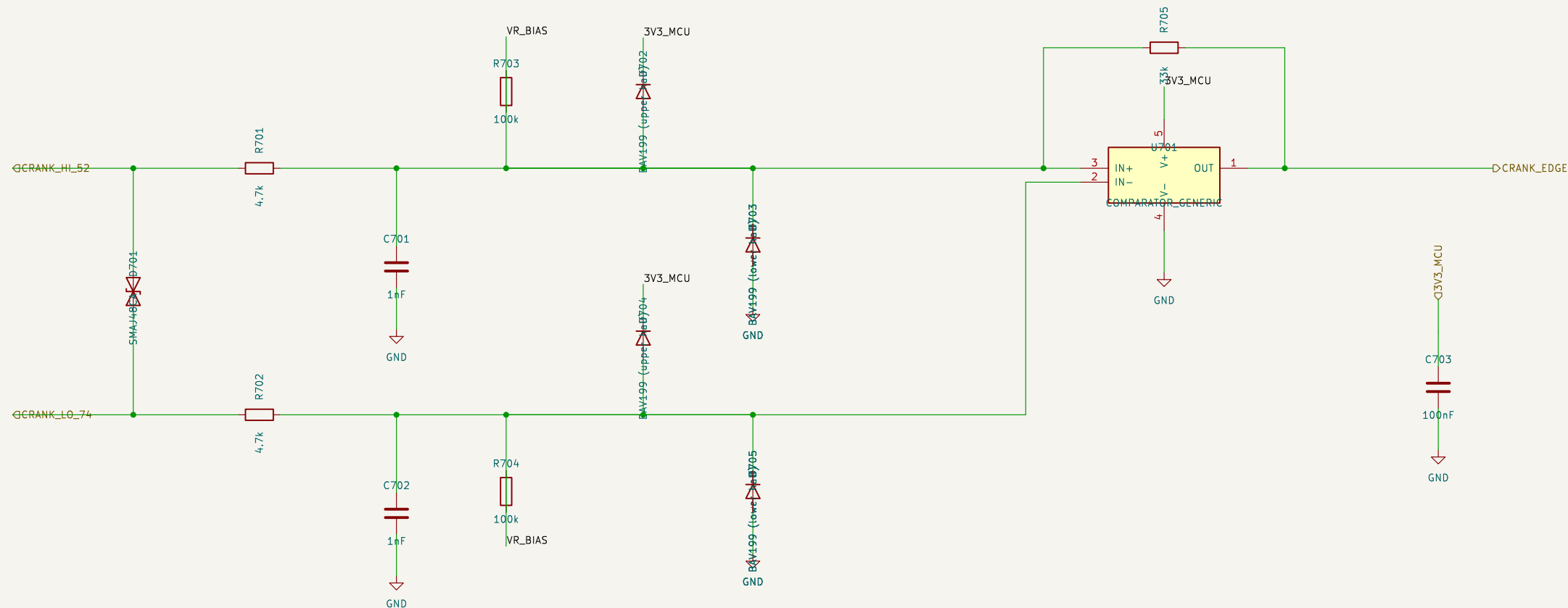
ECU pin 37 -- EGR position, return SHARED on pin 36 -> PTA15



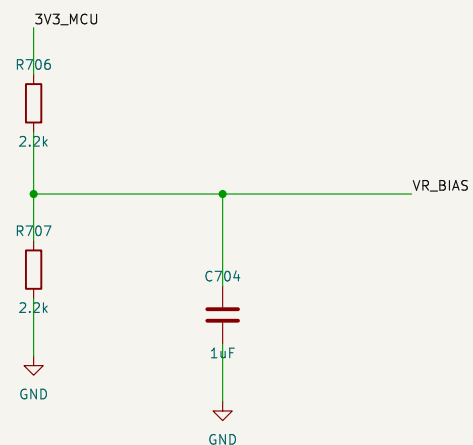
ECU pin 80 -- oil pressure, return SHARED on pin 36 -> PTA7



CRANK -- variable reluctance, ECU pins 52 / 74, shield 30 -> zero-cross comparator -> PTB2 (FTM1_CH0)



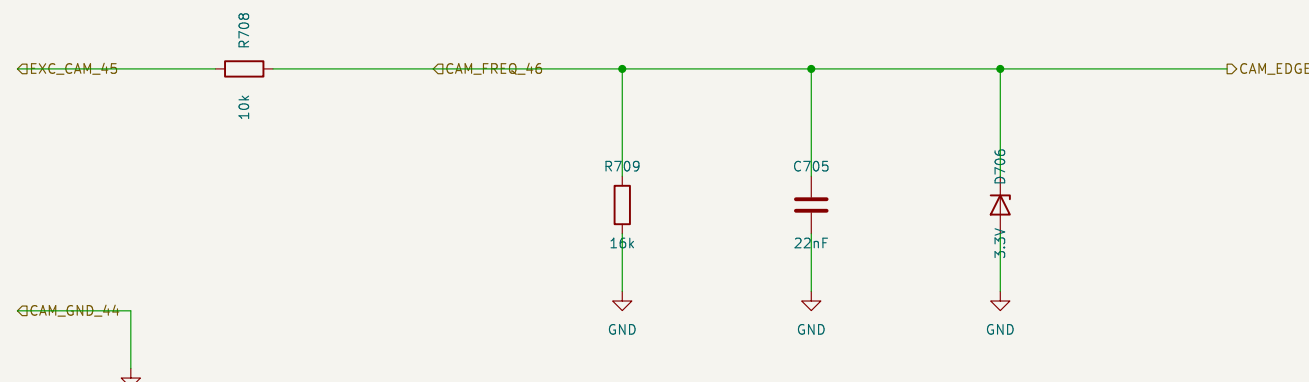
VR_BIAS -- mid-rail reference for the floating VR coil



ECU pin 30 -- crank sensor shield, terminated at THIS end only. Grounding both ends makes the shield a second conductor between two grounds that differ by the injector return current, and that current then flows down the shield around the pair it is there to protect.



CAM -- digital Hall, ECU pins 45 / 46 / 44 -> PTB3 (FTM1_CH1). CORRECTS the front-end spec sec.4 specifies literally



CLAMP ^{AND} RELEASE IS THE BINDING SPEC, WHICH INVERTS THE RULE THIS PROJECT USED ON THE POWER PATH. Every earlier clamp on this board was chosen for forward drop -- the negative-clamp Schottky's 0.547 V at 15 A decomposes into junction and bulk terms and that decomposition is what ruled out paralleling junction devices. Here the forward drop does not matter at all: these diodes conduct only on a fault, into a 4.7k series resistor that is already doing the limiting. What matters is what they do when they are OFF.

A reverse-biased diode's leakage flows out of the 4.489k Thevenin at the input node and appears as an offset voltage there. The hysteresis window is ± 198 mV, so:

BAT54S-class Schottky ~2 uA at 25 C, and Schottky leakage rises roughly 100x to 125 C -> ~200 uA -> 0.90 V of offset. That is 4.5x the whole hysteresis window. The comparator latches and the engine has no speed signal at temperature. BAV199-class low-leakage silicon 5 nA max at 25 C, under 1 uA at 125 C -> under 4.5 mV of offset, 2% of the window.

The two clamp diodes are biased symmetrically -- both reverse-biased by 1.65 V from V_{BIAS}) so their leakages partly cancel, and both legs are matched so what survives is partly common-mode -- but a design whose speed input depends on cancellation between two diodes' hot leakage is not one to defend. The requirement is stated instead: reverse leakage at or below 1 uA at 125 C. `vr conditioner.cir` does NOT model this -- its comparator is an ideal switch with no input network at all, so no clamp, no leakage and no offset exist in it. This is the sheet finding something the block could not.

WHY THE CAM FRONT-END ON THIS SHEET IS NOT THE ONE THE SPEC ASKS FOR. Spec sec.4 and docs/pinmap.md sec.1.3 both give pin 46's front end as "Pull-up to 5V_SENSOR -> timer capture": the Hall output pulled to 5 V and taken straight into PTB3. NXP S32K1xx Data Sheet Rev. 15, Table 17, puts Vih max at VDD + 0.3 V, about 3.6 V at VDD = 3.3 V, so a full 5 V swing is not a guaranteed valid logic level. Table 1 does allow -0.8 to +5.8 V on any I/O pin, but only while respecting the +/-3 mA injection limit, and the datasheet's own words are that "functional operation at the maximum values is not guaranteed" -- a stress margin, not a design basis for a pin that would sit at 5 V once per cam revolution for the machine's running life. The 16k to ground turns it into the same 10k/16k divider the ratiometric sensor channels already use. It costs one resistor.

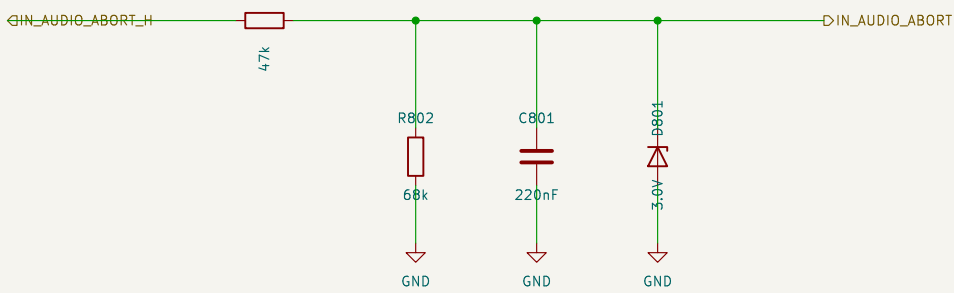
This was found by writing `cam_frontend.cir`, which `pinmap.sec4` asked for by name before this sheet was drawn: no block had ever simulated a cam front-end -- `vr_conditioner.cir` covers the crank and there was no cam equivalent -- so a 5 V swing on a 3.3 V pin had never been checked. Fourth finding of the same shape in this project: a document and an executable file disagreeing, with nothing comparing them.

PARTS NOT CHOSEN ON THIS SHEET: the comparator (COMPARATOR_GENERIC -- placeholder pins, no footprint) and the com.Hall sensor itself, which is UB on the unknowns register and needs the machine. The comparator's requirement is unusually easy, and deliberately so: biasing the floating coil to VR_BIAS means the inputs never leave mid-rail, so no part is needed whose common-mode range includes ground -- the usual hard constraint on a zero-cross detector. Single 3.3 v supply, push-pull rail-to-rail output, propagation delay at or below 1 us (0.009 canal degrees at rated speed on a 60-tooth wheel). Input offset voltage is a non-issue against a 198 mV window.

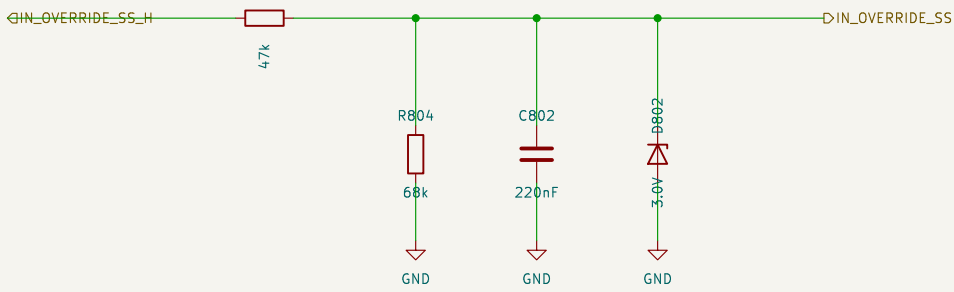
Clamp diodes specified by REVERSE LEAKAGE, not forward drop -- see the sheet note
 Zero-cross with +/-198 mV hysteresis: a fixed threshold cannot see a 2 V cranking signal
 vr_conditioner.cir / cam_frontend.cir
 Generated by hwgen_sheet_speed_inputs.py -- do not hand-edit until it is retired
ecu25kva
 Sheet: /speed_inputs/
 File: speed_inputs.kicad_sch

Title: Crank VR conditioner and cam front-end		Rev:
Size: A2	Date:	
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 8/11

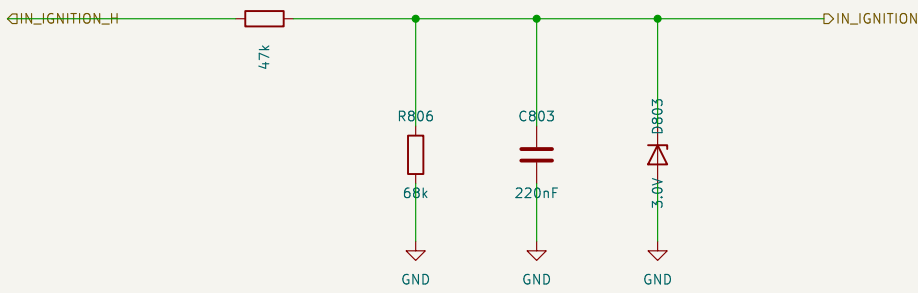
ECU pin 20 -- audio abort, active high -> PTD11



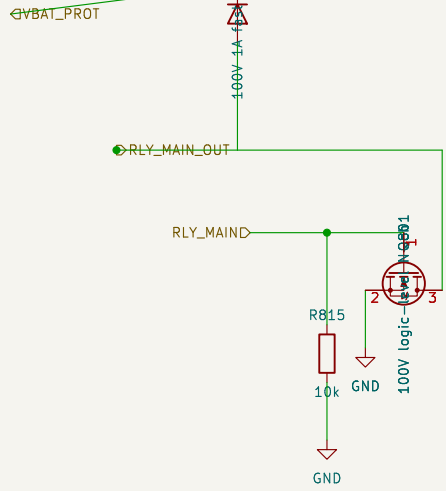
ECU pin 24 -- override start/stop, active high -> PTD12



ECU pin 71 -- ignition, active high -> PTD16



ECU pin 50 -- main relay, low side <- PTD13

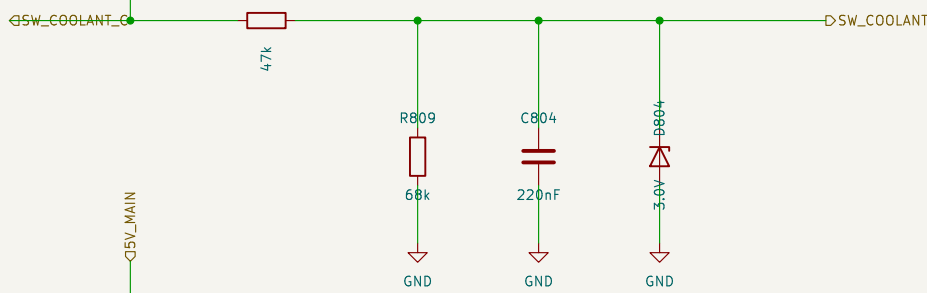


TRIP SENSE MOVES FROM PTD6 TO PTB0 (ADC0_SE4). docs/pinmap.md assigns it as a GPIO input; trip_module_sense.cir designs a THREE-STATE supervised loop and checks the three bands with real margins -- healthy 2.863-2.954 V, tripped 0.310-2.068 V, wire fault 0.000 V. A GPIO reads two states. It separates healthy from not-healthy and nothing else, which discards exactly the distinction the bleed resistor exists to create: a supervised loop that cannot report a cut wire is an unsupervised loop with extra parts. PTB0 is free, bonded out on the 144-LQFP, and Hi-Z with no pull at reset. PTD6 returns to the spare pool.

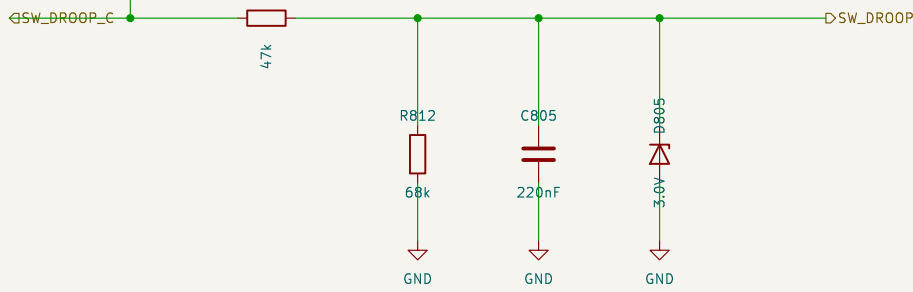
The 1.2 Mohm bleed resistor that makes the tripped band is INSIDE the trip module, across its contact -- it is not on this board and is not drawn here. It is also INFERRED rather than measured (trip_module_sense.cir says so), so the band edges above move if the real part differs.

WHY THE SAME FRONT-END ON EVERY CHANNEL. The 47k/68k/220nF/3.0V chain is discrete_input.cir's, and it is used unchanged on all six inputs -- including the two whose switch is a contact with real rather than a feed. Those two get a 10k pull-up BEFORE the 47k, so a battery short on the harness still meets the clamp chain rather than the pull-up. An open contact then leaves the divider at 2.72 V, under the 3.0 V knee, so it reads high without the zener conducting.

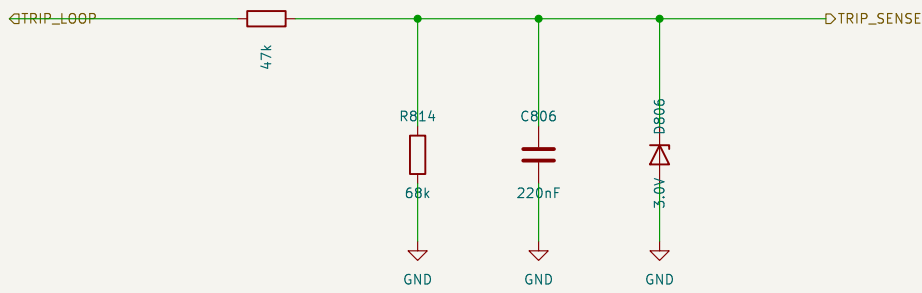
ECU pin 43 -- coolant switch, contact to ground -> PTD5



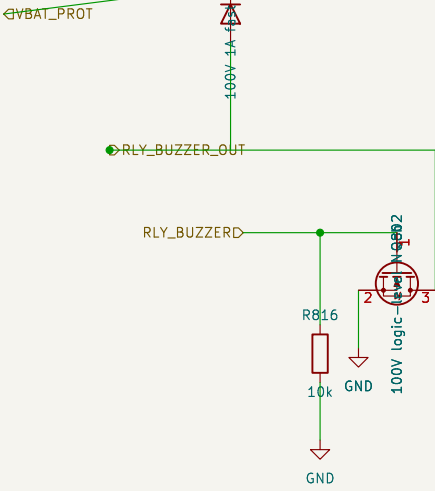
ECU pin 23 -- droop switch, contact to ground -> PTD7



NEW pin -- supervised trip loop -> PTB0 (ADC0_SE4), not PTD6



ECU pin 69 -- buzzer relay, low side <- PTD14



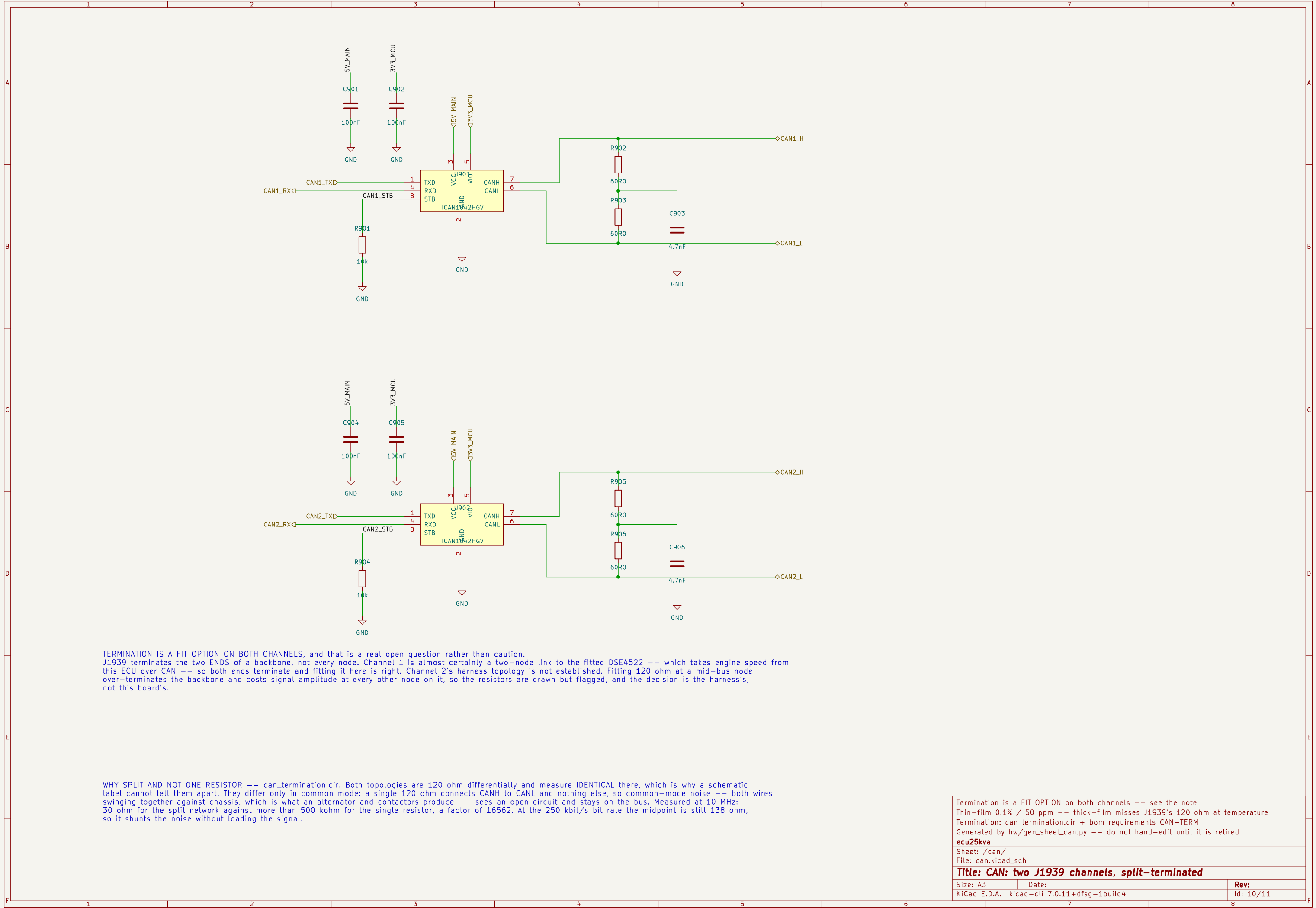
Relay FETs are 100 V class -- VBAT_PROT reaches 73.3 V on pulse 2a
Trip sense moved PTD6 -> PTB0: a GPIO cannot read a three-state loop
Front-ends: discrete_input.cir / trip_module_sense.cir. Outputs: relay_driver.cir
Generated by hw/gen_sheet_discrete_io.py -- do not hand-edit until it is retired

ec25kva

Sheet: /discrete_io/
File: discrete_io.kicad_sch

Title: Discrete I/O: switch inputs, trip loop, relay drivers

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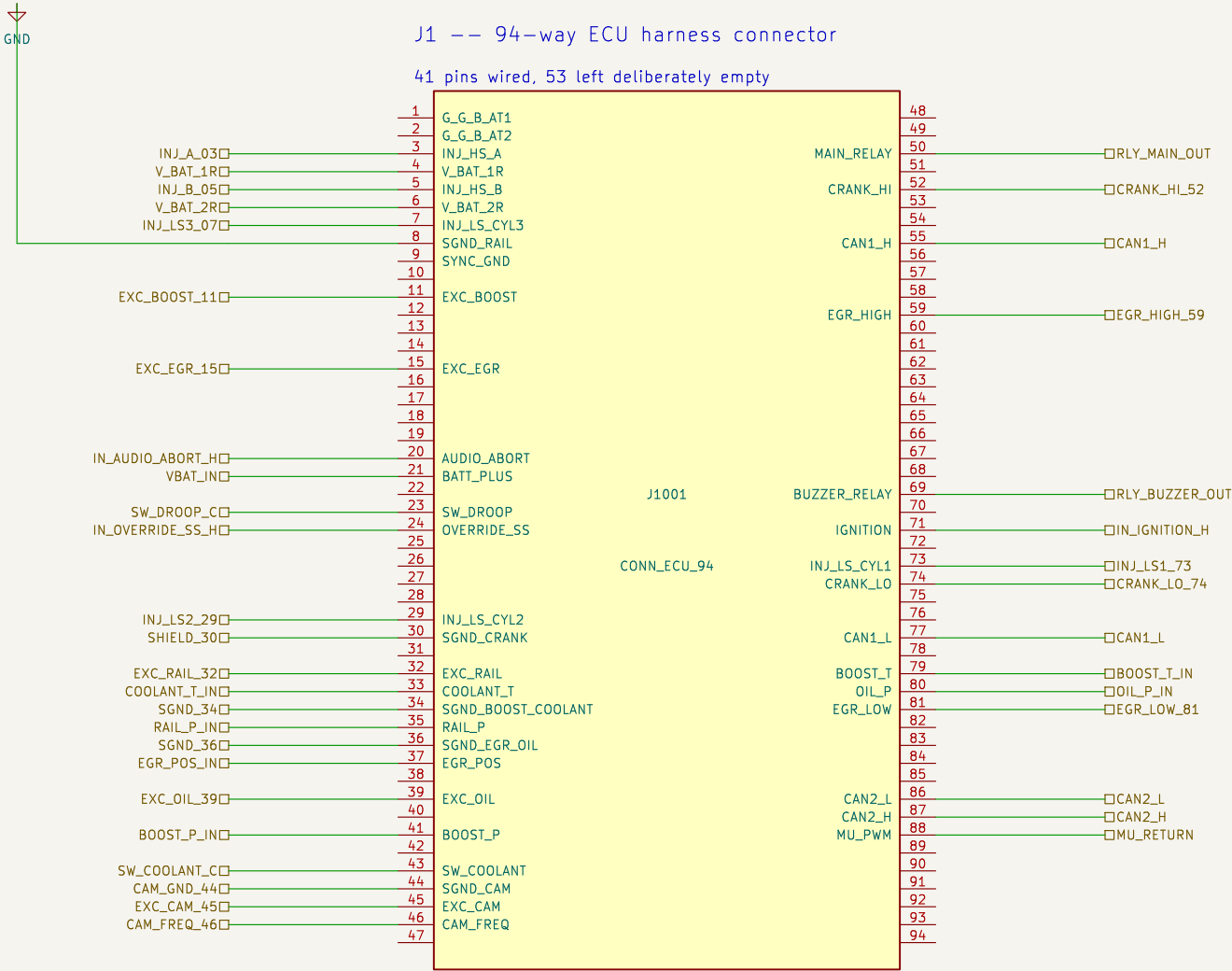
THE POWER GROUND HAS NO PIN, AND THAT IS THIS SHEET'S FINDING.

GP3.8703.C4.pdf shows pin 21 as the battery positive feed and shows no battery negative anywhere on the ECU connector. Its own scope note explains why — it is a field troubleshooting diagram and it omits "power grounds, unused pins and internal-only pins" — so this is a gap in the SOURCE, not a contradiction in it.

It is still a gap that stops a board. Every other sheet in this project returns to GND: the injector low sides carry 18 A into it, the metering unit up to 1.35 A, and the analog front ends reference their zeners and their 22 nF filters to it. At this boundary that net has nowhere to go.

It cannot be solved by choosing. Fifty pins have no established function and several of them are certainly grounds — a 94-way connector feeding an 18 A injector stage does not have one return — but picking which is exactly the guess this project forbids, and a wrong ground pin is not a rework, it is a harness that has to be remade.

WHAT ACTUALLY SETTLES IT: the connector part number and the OEM pin list, or continuity measurement on the machine between the connector shell and battery negative. The reference names the first as the missing document. The second needs the machine and belongs with U16/U2/U8/U14/U5.



PINS SHOWN ON THE DIAGRAM AND STILL NOT WIRED

- 09 SYNCHRONIZATION GROUND. Drawn RED on a diagram whose own legend makes red a positive feed, so the name and the colour disagree. U6 on the unknowns register.
- 01 G_G_B AT1. 02 G_G_B AT2. Memo 03 suspects these are power grounds rather than switch-return legs. If that is right they are part of the answer to the note on the left. No MCU pin is assigned either way, because guessing which would violate the same rule.

NETS THAT EXIST AND HAVE NO PIN YET

TRIP_LOOP the trip-module supervised loop is NEW — it is not on the OEM diagram at all, so it needs one of the 50 unconfirmed pins. discrete_io draws the front end; the pin waits on the same document the ground does.

EGR_HIGH_59 wired here, because pins 59 and 81 ARE on the diagram — but the bridge that drives them is the blocked half of metering_egr, waiting on a gate driver rated above the 73.3 V this board's own battery rail reaches on ISO 7637-2 pulse 2a. The pins are real; the driver is not chosen.

WHY PINS 34 AND 36 ARE SIGNALS AND PIN 08 IS GROUND

34 and 36 are SHARED sensor returns — 34 for boost pressure and coolant temperature, 36 for EGR position and oil pressure. A shared return carries both sensors' current, so its own resistance is an offset on both readings, which is why sensors_analog carries them into difference amplifiers rather than to ground. 08 is rail pressure's DEDICATED return: nothing to reject, so it is simply ground here.

NO PART NUMBER, SO NO FOOTPRINT. The reference records a 94-pin connector from GP3.8703.C4.pdf page 4 and nothing further — no Bosch part number, no keying, no mating half. CONN_ECU_94 therefore carries pin COUNT and pin NUMBERS, which are facts read off a document, and no footprint, which would be a guess about a physical interface that has to mate with a harness already on a machine. Pin NAMES on the symbol are the diagram's own signal names where it shows the pin, and KiCad's blank elsewhere. This is the same treatment the op-amps on sensors_analog and the comparator on speed_inputs get, for the same reason: the symbol asserts what is known and declines to assert what is not.